

Jin Woo Oh

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SKILLS

Languages: Python, SQL, TypeScript (React), JavaScript, R, C++

Machine Learning & AI: LLM (Claude, GPT), generative AI (GenAI), Claude Code (custom skills, hooks, MCP), prompt engineering, vLLM (self-hosted inference)

Cloud & DevOps: AWS, GCP, Kubernetes, Docker, Modal, CI/CD (GitHub Actions), Temporal, monitoring & alerting

Data & API: PostgreSQL, NoSQL, BigQuery, Power BI, REST API, microservices, FastAPI, PostHog

EXPERIENCE

Avnir **Nov 2025 – Present**
Software Engineer *Atlanta, GA*

- Developed location-intelligence microservice end-to-end (REST API with OAuth and React/TypeScript UI), serving knowledge graph-based recommendations on PostgreSQL's Apache AGE, exposed via MCP server for agentic AI workflows
- Built LLM-powered document-processing pipeline with Claude prompt engineering and concurrent Temporal workflows, cutting data ingestion from 40 to 4 minutes
- Engineered relationship-strength scoring neural network model with 30 behavioral features using scikit-learn grid search and stratified k-fold cross-validation, optimizing scoring computation from 75 minutes to 5 seconds
- Migrated 3 production GenAI microservices (vLLM inference servers) from Qwen2.5 to Gemma 4 on serverless GPUs (Modal), cutting compute costs 59% and eliminating cold-start failures
- Diagnosed 45-minute production LLM pipeline stall via root-cause analysis; added KEDA autoscaling, queue-backlog monitoring and alerting, ensuring high availability of mission-critical ingestion
- Automated KPI reporting via 30-chart Apache Superset dashboards, delivering weekly business insights to non-technical stakeholders and anomaly monitoring for engineers, saving 12+ hours of manual reporting weekly

ICC Lab, Florida State University **May 2025 – Nov 2025**
Researcher *Tallahassee, FL (Remote)*

- Built multimodal computer-vision pipeline extracting structured gaze data from Tobii eye-tracker video with OpenCV, automating 1,000+ hours of manual labeling
- Achieved 95% accuracy in detecting UI state changes via robust template matching and HSV color masking

FlightAware **June 2024 – Aug 2024**
Software Engineer Intern *Houston, TX (Remote)*

- Reduced deep-learning application latency 35% by creating scalable Python API serving model inference across internal teams
- Evaluated conversational AI chatbots (GPT-4o vs. Mistral-7B RAG) through A/B testing, recommending GPT-4o for 21x faster responses and 40% accuracy gains

Samsung Electronics America **Aug 2019 – Apr 2024**
Software Quality Engineer III *Berkeley Heights, NJ*

- Spearheaded GPU-accelerated NLP classifier detecting critical flagship-device complaints at 92% accuracy, automating 11+ hours of weekly manual triage
- Automated ETL pipeline (Python, SQL) processing 1M+ articles for daily QA reports, accelerating issue detection 25%
- Reduced live-chat operating costs 30% via prescriptive discrete-event simulation modeling and statistical optimization of staffing allocation
- Presented actionable business insights to technical and non-technical stakeholders through user-behavior and QA trend dashboards in Power BI

PROJECTS

Technical Issue Classifier | PyTorch, HuggingFace, Docker, Kubernetes

- Optimized NLP issue classifier decision threshold via ROC analysis and grid search, reducing costly false negatives to 3%

RayTracing | C++, CUDA, GPU acceleration

- Implemented CUDA ray tracer achieving 35x GPU speedup over CPU via parallel programming and HPC scaling

EDUCATION

Georgia Institute of Technology **Expected Dec 2026**
Master of Science (M.S.) in Analytics *Atlanta, GA (Online)*

Seoul National University **Feb 2018**
Master of Science (M.S.) in Mechanical Engineering *Seoul, Korea*

- 2 patents, 2 journal publications, 8 conference presentations; Academic Scholarship; 3.75/4.0 GPA, *Summa cum laude*

University of Illinois at Urbana-Champaign **May 2015**
Bachelor of Science (B.S.) in Mechanical Engineering *Champaign, IL*

- 1 journal publication; VP, Co-founder, mentor and tutor at Korean Mechanical Engineer Network; Academic Scholarship